Predicting Online Transaction Fraud Using Machine Learning

Preliminary Analysis Report

Date: March 2026

1. Dataset Overview

The IEEE-CIS Fraud Detection dataset provided by Vesta Corporation was loaded and merged from two files, train_transaction.csv and train_identity.csv, joined on TransactionID. The merged dataset contains 590,540 transactions and 434 features. The fraud rate is 3.5%, confirming a severe class imbalance consistent with real-world fraud data.

2. Data Cleaning

Columns with more than 70% missing values were dropped, removing 208 columns and reducing the feature set to 226. Remaining missing values were filled using column medians. Categorical variables were encoded using Label Encoding.

3. Feature Engineering, Severity Tier

A SeverityTier feature was engineered from TransactionAmt to segment fraud by financial impact:

Severity Tier	Transaction Amount	Transaction Count
Low	< $50	187,515
Medium	$50 – $500	380,146
High	> $500	22,879

This is our key differentiator from existing work on this dataset — most solutions only predict whether fraud occurs, not how damaging it is.

4. Class Imbalance Handling

Two techniques were applied and compared:

- **SMOTE** (Synthetic Minority Oversampling Technique): Applied only to training data, oversampling fraud cases to 16.67% of training set (targeting 20%). After SMOTE, training size increased from 413,378 to 478,696.
- **scale_pos_weight = 27.58**: Applied in XGBoost to weight fraud cases ~28x more heavily than legitimate transactions.

5. Model Results

Model	**AUC-ROC**
Logistic Regression	0.6478
Random Forest	0.9196
XGBoost with SMOTE	0.9074
XGBoost with scale_pos_weight	**0.9260**

XGBoost with scale_pos_weight achieved the highest AUC-ROC of 0.9260, falling within the anticipated range of 0.90–0.94 stated in our proposal. Logistic Regression confirms the problem requires non-linear models. XGBoost with scale_pos_weight marginally outperformed SMOTE, suggesting native imbalance handling is more effective for this dataset.

6. Severity Tier Performance (Our Differentiator)

Severity Tier	**Fraud Cases**	**AUC-ROC**	**Recall**
Low (<$50)	2,239	0.9261	0.5033
Medium ($50–$500)	3,653	0.8982	0.3384
High (>$500)	**307**	**0.8644**	**0.1629**

Key Finding: Model performance consistently drops as transaction severity increases. Highvalue frauds — which cause the greatest financial harm — have the lowest recall at 16.3%. This means the model currently misses over 80% of the most damaging fraud cases. This directly motivates threshold optimization weighted by transaction amount in our next steps.

7. Confusion Matrix (XGBoost with SMOTE at threshold = 0.5)

	Predicted Legit	**Predicted Fraud**
Actual Legit	170,520 (TN)	443 (FP)
Actual Fraud	3,786 (FN)	2,413 (TP)

At the default threshold of 0.5, the model is very conservative, flagging very few transactions as fraud. The 3,786 false negatives represent missed frauds causing real financial harm.

8. Recommended Classification Threshold

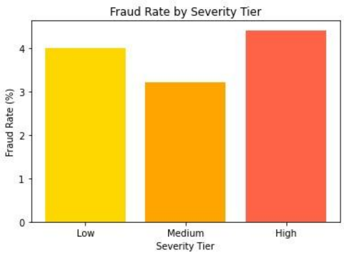
- **Recommended Threshold:** 0.2882
- **Precision:** 0.6845
- **Recall:** 0.5093
- **F1-Score:** 0.5840

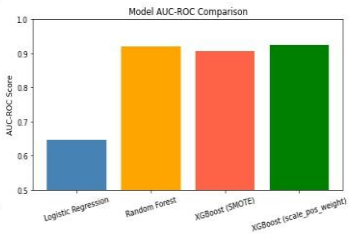
Lowering the threshold from 0.5 to 0.2882 significantly improves recall, catching more fraud at the cost of some precision. For real-world deployment, this trade-off is preferable since missing fraud is more costly than a false alert.

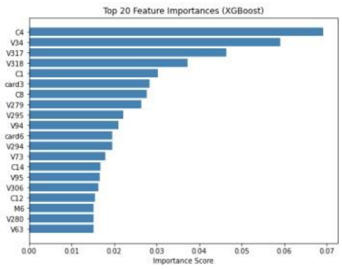
9. Geography Proxy, Email Domain Analysis

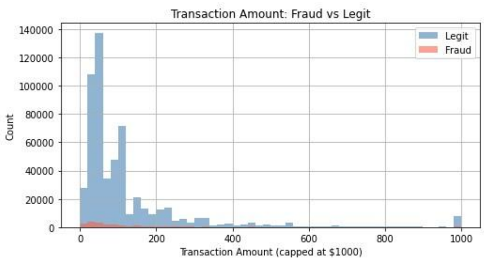
Fraud rates vary significantly across email domains. Outlook.com shows the highest fraud rate at approximately 9.4%, nearly double hotmail.com at 5.3% and gmail.com at 4.4%. Lower-traffic domains such as att.net show minimal fraud activity. Email domain is confirmed as a strong behavioral signal for fraud risk.

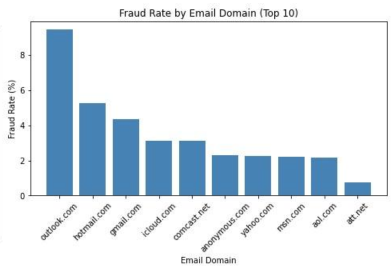
10. Visualizations:

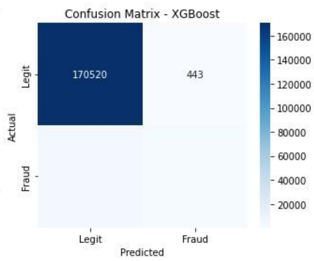


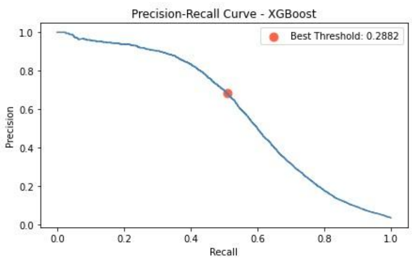












11. Next Steps

- Feature selection to identify top 15–20 predictors from 224 remaining features
- Deep analysis of V-series anonymized features using importance scores
- Cost-sensitive threshold optimization weighted by TransactionAmt
- Cross-validation and final model tuning
- Severity-weighted evaluation metric to penalize missed high-value fraud more heavily